Sign Language Detection System

Phase 3 — Work Done Report

Project Title	Sign Language Detection System
Phase	Phase 3
Document Type	Work Done Report (Individual Contributions)
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1. Purpose of this Document

This report documents the work completed by each team member during the Phase 3 reporting period of the Sign Language Detection System project. It covers not only the core Phase 3 deliverables but also related activities carried out during this period, including research, learning, experimentation, dataset-related work, meetings, and preliminary implementation, in order to give an accurate and complete picture of each member's individual contribution.

2. Team Overview

Team Member	Phase 3 Responsibility
Archana Sudhakaran	System Architecture
Shafna K N	Database Architecture
Akshara M	UI/UX Design
Karthik Prasad	System Diagrams (Architecture, ER, Use Case)

3. Individual Contributions

Archana Sudhakaran — System Architecture

Responsible for defining the overall system architecture of the Sign Language Detection System, ensuring that all modules (data input, model inference, and UI/output layers) integrate into a coherent, scalable design.

- Designed the end-to-end system architecture covering the video/image capture pipeline, preprocessing module, model inference layer, and result output layer.
- Defined the interaction and data flow between the frontend, backend, and the sign language recognition model.
- Prepared the Architecture Diagram (Architecture_Diagram.png) illustrating component boundaries, data flow, and integration points.
- Researched suitable architectural patterns (client-server vs. modular pipeline) for real-time gesture recognition systems.
- Evaluated technology choices for deployment (local inference vs. API-based model serving) and documented trade-offs.

- Participated in team meetings to align system architecture decisions with the database design and UI/UX workflows.
- Reviewed integration points with the database and UI modules to ensure architectural consistency across the project.

Shafna K N — Database Architecture

Responsible for designing the data storage structure required to support user data, session logs, detection history, and model-related metadata for the Sign Language Detection System.

- Designed the database schema covering entities such as users, detection sessions, recognized signs/gestures, and feedback/logs.
- Prepared the Entity-Relationship Diagram (ER_Diagram.png) defining entities, attributes, and relationships.
- Normalized the schema to reduce redundancy and maintain data integrity across detection records.
- Researched appropriate database technology (relational vs. NoSQL) suited to storing sequential gesture/detection data.
- Defined keys, constraints, and indexing strategy to support efficient querying of detection history.
- Collaborated with the system architecture and UI/UX members to ensure the schema supports required application features.
- Attended team meetings to review and finalize the data model against project requirements.

Akshara M — UI/UX Design

Responsible for designing the user interface and overall user experience of the Sign Language Detection application, focusing on usability and accessibility.

- Designed wireframes and mockups for the core screens: sign capture/detection view, results display, and history/feedback screens.
- Defined the visual design language (colour scheme, typography, iconography) for the application.
- Conducted usability considerations research for accessibility, given the application's target users.
- Mapped user flows from sign capture to detection output, ensuring a smooth end-to-end experience.
- Coordinated with the system architecture lead to ensure UI designs are feasible within the proposed system flow.
- Participated in design review meetings and incorporated team feedback into revised UI mockups.

Karthik Prasad — System Diagrams — Architecture, ER, and Use Case

Responsible for consolidating and formalizing the project's core technical diagrams, translating design decisions from the architecture, database, and UI teams into standard diagrammatic representations.

- Prepared the Use Case Diagram (Use_Case_Diagram.png) identifying key actors (user, system/model) and use cases such as sign capture, gesture recognition, and result viewing.
- Assisted in refining and finalizing the Architecture Diagram (Architecture_Diagram.png) in coordination with Archana Sudhakaran.
- Assisted in refining and finalizing the ER Diagram (ER_Diagram.png) in coordination with Shafna K N.
- Researched standard UML/ER notations and diagramming best practices to ensure consistency across all three diagrams.
- Cross-checked diagrams against one another to ensure consistency between the system architecture, data model, and use cases.
- Participated in team meetings to walk through and validate all diagrams before finalization.

4. Team-Wide Activities

The following activities were carried out collectively by the team during Phase 3, in addition to individual responsibilities:

- Conducted periodic team meetings to track progress, resolve integration issues, and align on design decisions across architecture, database, and UI/UX.
- Carried out preliminary research into existing sign language detection approaches and datasets relevant to the project.
- Reviewed and discussed candidate datasets for gesture/sign recognition to inform system and database design choices.
- Cross-reviewed each other's diagrams and deliverables (architecture, ER, use case) to ensure consistency across the system design.
- Documented open questions and dependencies to be carried forward into the next phase (e.g., model selection, dataset finalization).

5. Summary

During Phase 3, the team made concrete progress on the foundational design of the Sign Language Detection System. Archana Sudhakaran established the system architecture, Shafna K N designed the database architecture, Akshara M produced the UI/UX design, and Karthik Prasad consolidated the architecture, ER, and use case diagrams. Alongside these individual deliverables, the team engaged in shared research, dataset review, and coordination meetings to ensure all components of the design remain consistent and ready for implementation in the next phase.